

Hands-On-Learning: Responsible AI Presentation Challenge

Learning Objectives

By the end of this hands-on practice, you will be able to:

- Transform Responsible AI documentation into clear, visual presentations for non-technical audiences
- Use Python to generate presentation slides programmatically
- Apply storytelling techniques to highlight fairness, transparency, and governance in AI design
- Reflect on how clarity and trust enhance stakeholder engagement in Responsible AI discussions



Description

In this hands-on Visual Studio Code practice, you'll take on the role of the Responsible AI Lead at MoveLogix, a logistics company launching a route-optimization algorithm. Executives have requested a concise, non-technical presentation showing how your design mitigates ethical risks—such as unfair route assignments, data misuse, and lack of transparency—while preserving operational efficiency. Using Python and the python-pptx library, you'll generate a short stakeholder-ready pitch deck that communicates fairness, accountability, and trust clearly and persuasively.

This hands-on practice focuses on the communication aspect of Responsible AI. You'll create a structured JSON file describing your system's ethical safeguards and generate presentation slides directly from code to ensure consistency, accuracy, and traceability. By combining storytelling principles with automation, you'll learn how to translate technical AI ethics work into accessible narratives that inspire confidence among leadership and regulators alike.

Structure of the Activity

Dataset Information

Filename: presentation_data.json

Format: Structured JSON containing presentation content

Key Features: System overview and purpose, identified ethical risks and mitigation notes, key transparency and oversight points

Tools Used:

- Visual Studio Code (any recent version)
- Python 3.10 or higher
- **Required libraries:** python-pptx (for PowerPoint generation)

Setup Requirements

Technical Prerequisites: Python 3.10+ installed and accessible from command line

Library Installation: Run `pip install python-pptx` before starting
System Requirements: Any operating system with Visual Studio Code support

Required Files in Working Directory: `presentation_data.json`, `speaker_notes.txt`, `reflection.md` (all created during the hands-on practice)

Scenario

You are the Responsible AI Lead at MoveLogix, a logistics company preparing to launch a route-optimization algorithm that assigns delivery routes to drivers. Executives have requested a concise, non-technical presentation demonstrating how your design mitigates ethical risks—including unfair route assignments that could disadvantage certain drivers, potential data misuse, and lack of transparency in algorithmic decisions. Your presentation must communicate fairness, accountability, and trust to secure leadership approval before launch.

Objective

Create a structured data file for presentation content, programmatically generate a 6–8 slide pitch deck using Python, write speaker notes for non-technical delivery, and reflect on effective AI responsibility communication.



Getting Started

- 1 Open Visual Studio Code and create a new folder for your hands-on practice work
- 2 Install the required library by running: `pip install python-pptx`
- 3 Create a new file named `presentation_data.json` for your slide content
- 4 Create a new file named `speaker_notes.txt` for your delivery notes
- 5 Create a new file named `reflection.md` for your written reflection
- 6 Verify `python-pptx` installation by running: `python -c "from pptx import Presentation; print('Ready!')"`

Instructions

Activity 1: Prepare Your Responsible AI Content

Objective

Create a structured JSON file containing all content for your Responsible AI presentation, organized for programmatic slide generation.



Input/Data

Your presentation_data.json file (initially empty)

Context: MoveLogix route-optimization algorithm assigns delivery routes based on driver location, package volume, and delivery windows

Steps to Complete

Step 1: Design a JSON structure with sections for each planned slide. Include keys for: system_overview, ethical_risks, mitigations, transparency_measures, governance_highlights, and next_steps.

Step 2: Write the `system_overview` section describing MoveLogix's route-optimization algorithm purpose and business value. Use accessible, human-centered language that executives can understand.

Step 3: Document at least 3 `ethical_risks` relevant to route optimization (e.g., unfair route assignments to certain driver demographics, privacy concerns with location tracking, lack of explainability for route decisions).

Step 4: Write corresponding mitigations for each risk, focusing on concrete actions the team has implemented or will implement.

Step 5: Add `transparency_measures` explaining how the system's decisions can be understood and audited, and `governance_highlights` describing oversight processes.

Expected Output

A valid JSON file with structured content for 6–8 slides

Content written in accessible, non-technical language

At least 3 risks with corresponding mitigations

Tips

Write content as if explaining to someone without AI expertise

Keep bullet points concise—slides should have minimal text

Use action verbs for mitigations (e.g., "Monitor," "Audit," "Review")

Activity 2: Generate and Annotate Your Slides

Objective

Build a Python script that reads your JSON content and generates a professional PowerPoint presentation programmatically, then write speaker notes for delivery.



Input/Data

Your completed presentation_data.json from Activity 1

Required imports for presentation generation:

```
from pptx import Presentation
from pptx.util import Inches, Pt
import json
```

Steps to Complete

Step 1: Create a Python script that loads your presentation_data.json file using the json module.

Step 2: Initialize a new Presentation object and implement a function to add slides. Use slide layouts for title slides (layout index 0) and content slides (layout index 1 for title + content).

Step 3: Generate slides for each section of your JSON content. Create slides for: Title/Purpose, Ethical Risks, Mitigations, Transparency Measures, Governance, and Next Steps (6–8 slides total).

Step 4: Add content to each slide by accessing the title and body placeholders. For bullet points, add paragraphs to the text frame of the content placeholder.

Step 5: Save the presentation as ResponsibleAI_PitchDeck.pptx using the Presentation.save() method.

Step 6: Open `speaker_notes.txt` and write delivery notes for each slide explaining the key points in plain, non-technical language suitable for executive presentation.

Expected Output

A Python script that successfully generates a .pptx file

`ResponsibleAI_PitchDeck.pptx` with 6–8 professionally formatted slides

`speaker_notes.txt` with delivery guidance for each slide

Tips

Access slide placeholders using `slide.shapes.title` and `slide.placeholders[1]` for body content

Add bullet points by calling `text_frame.add_paragraph()` for each item

Test your script after adding each slide to catch errors early

Speaker notes should expand on slide content, not repeat it verbatim

Activity 3: Reflect on Communicating AI Responsibility

Objective

Write a reflection on how effective communication builds stakeholder trust and demonstrates accountability in Responsible AI initiatives.



Input/Data

Your reflection.md file

Your completed presentation and speaker notes from Activities 1 and 2

Steps to Complete

Step 1: Open your reflection.md file and create a heading for your reflection

Step 2: Write approximately 200 words addressing how effective communication builds stakeholder trust in AI systems

Step 3: Discuss the role of clarity and accessibility in demonstrating accountability to non-technical audiences

Step 4: Reflect on the value of programmatic presentation generation for consistency and traceability in Responsible AI documentation

Expected Output

A markdown file with approximately 200 words of reflective writing

Clear connection between communication quality and stakeholder trust

Insight into the benefits of code-generated documentation

Tips

Consider how the same technical content might be perceived differently based on presentation quality

Think about how automation reduces human error and ensures content-presentation alignment

Reference specific choices you made in your presentation design

Compile Your Findings

Requirement 1: Presentation Data Source

Completed presentation_data.json file

Include:

- System overview section with accessible description
- At least 3 documented ethical risks
- Corresponding mitigations for each risk
- Transparency and governance sections
- Valid JSON syntax

Requirement 2: Generated Presentation

ResponsibleAI_PitchDeck.pptx and Python generation script

Include:

- 6–8 slides covering Purpose, Risks, Mitigations, Transparency, and Next Steps
- Professional formatting with clear titles and bullet points
- Python script that reads JSON and generates the presentation
- Speaker notes in speaker_notes.txt for non-technical delivery

Requirement 3: Communication Reflection

Completed reflection.md file

Include:

- Approximately 200 words of reflective analysis
- Discussion of communication's role in building stakeholder trust
- Insight into benefits of programmatic documentation generation

Summary

This activity teaches Responsible AI communication through practical implementation of automated presentation generation. You've learned how to translate technical AI ethics documentation into accessible stakeholder narratives, use Python to ensure consistency and traceability in deliverables, and apply storytelling principles to build trust—skills directly applicable in AI product roles, ethics consulting, and any position requiring executive-level communication of technical governance topics.

Share The Project

Document: Save the report in PDF format.

Upload: Share in the “Peer Review” area of the course shell with a brief description.

Instructions for Documenting and Sharing The Project for Peer Review

1. Document the Project

- Use word processing software to create your report.
- Ensure all sections of the project document structure are completed.
- Proofread and edit your report for clarity and accuracy.

2. Share the Project

- Save your report in PDF format.
- Upload your documents to the “Peer Review” area in the course shell.
- Provide a brief description of your project in the submission post.

Instructions for Documenting and Sharing The Project for Peer Review

3. Peer Review

- Review the projects submitted by your peers.
- Provide constructive feedback and comments on their work.